THE LARGEST m WITH $m!$ DIVIDING $\prod_{j=0}^{k}\binom{k}{j}$ EQUALS k IF AND ONLY IF $k+1$ IS PRIME: KARTTUNEN'S CONJECTURE FOR OEIS A249151 AND A249434

MATH0

Abstract. For $k \geq 0$ let $P(k) = \prod_{j=0}^{k}\binom{k}{j}$ be the product of the entries of row k of Pascal's triangle, and let $M(k)$ be the largest integer m such that $m!$ divides $P(k)$ (OEIS A249151). We prove that $M(k) = k$ if and only if $k+1$ is prime; equivalently, $k! \mid P(k)$ and $(k+1)! \nmid P(k)$ exactly when $k+1$ is prime. This settles a conjecture recorded by Antti Karttunen in 2014 in the OEIS entries A249434 and A249151, which still list it as conjectural. The proof is elementary: using $P(k+1)\,(k+1)! = P(k)\,(k+1)^{k+1}$ and a q-adic recursion for $v_q(P(n))$, we show that for every prime $q \mid k+1$ with $q \neq k+1$, $v_q(P(k)) - v_q(k!)$ is either negative or at least $v_q(k+1)$. To our knowledge the result is new. It is verified in Lean 4 against a frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the listed OEIS terms.

1. Introduction

For $n \geq 0$ let $P(n) = \prod_{j=0}^{n}\binom{n}{j}$ be the product of the entries of row n of Pascal's triangle. This is OEIS A001142 [9], $1, 1, 2, 9, 96, 2500, 162000, \ldots$. Lagarias and Mehta [5] studied the size and the prime factorization of $P(n)$ (their G_n, the reciprocal of the product of all nonzero unreduced Farey fractions of order n). They gave three formulas for the p-adic valuation $\mathrm{ord}_p(G_n)$, two of them in terms of base-p expansions, determined its extreme values, and showed in particular [5, Theorem 6.7] that for $n \geq 1$, $\mathrm{ord}_p(G_n) = 0$ if and only if $n = ap^c - 1$ with $1 \leq a \leq p-1$. Partial factorizations of G_n, and of products of extended binomial coefficients, were studied by Du and Lagarias [2] and by Du, Lagarias and Yangjit [3].

In October and November 2014 Antti Karttunen added to the OEIS the sequence A249151 [11], whose term $M(n)$ $(n \geq 0)$ is the largest m such that $m!$ divides $P(n)$, namely $1, 1, 2, 1, 4, 2, 6, 1, 2, 4, 10, 7, 12, 6, 4, 1, 16, \ldots$, together with A249434 [14], the integers $m \geq 0$ with $m! \mid P(m)$, that is, with $M(m) \geq m$, and A249429 [12], the integers $n \geq 0$ with $M(n) > n$. For example $M(4) = 4$, because $P(4) = 96 = 4 \cdot 4!$ and $5! \nmid 96$, while $M(3) = 1$ because $P(3) = 9$. The data suggest that $M(k) = k$ happens exactly when $k+1$ is prime, and Karttunen recorded this as a conjecture in A249434, A249151 and A249431 [13]. We prove it.

Theorem 1. *For every integer $k \geq 0$, $M(k) = k$ if and only if $k+1$ is prime. Equivalently, $k! \mid P(k)$ and $(k+1)! \nmid P(k)$ if and only if $k+1$ is prime.*

Method. Fix a prime q and put $n = k+1$. The identity $P(n)\,n! = P(n-1)\,n^n$ gives $v_q(P(k)) - v_q(k!) = D_q(n) := v_q(P(n)) - (n-1)v_q(n)$ (Lemma 4), so $k! \mid P(k)$ means $D_q(n) \geq 0$ for all q, and $(k+1)! \mid P(k)$ means $D_q(n) \geq v_q(n)$ for all q. A q-adic recursion for $v_q(P(n))$ (Lemma 7) yields a closed form for $D_q(q^e u)$ and an inequality between $v_q(P(u))$ and the base-q digit sum of u (Lemma 9). Together they show that if $q \mid n$ and $n \neq q$, then $D_q(n) < 0$ or $D_q(n) > v_q(n)$ (Proposition 11), while $D_q(q) = 0 < 1 = v_q(q)$.

Date: October 25, 2026.

Status and prior work. The OEIS entries A249434 (revision #20, September 10, 2025), A249151 (#38, August 24, 2025) and A249431 (#16, August 24, 2025) still state the claim with "It seems"; A249434 lists A006093 among its subsequences as "(conjectured)", and none of the entries records a proof. The vanishing criterion of Lagarias and Mehta [5, Theorem 6.7] describes the case $v_q(P(u)) = 0$ that occurs in our dichotomy (Remark 12), but [5] does not discuss the divisibility of G_n by factorials, the sequence A249151, or the primality of $n+1$, and the papers [2, 3] do not address the conjecture either. The conjecture is not among the OEIS statements of the formal-conjectures repository [4] (checked at the commit pinned by math0, Section 4), nor among the 492 OEIS conjectures formalized in Lean by Tsoukalas et al. [8] that make up the OEIS Open benchmark [1]. The math0 referees' searches of MathOverflow, Mathematics Stack Exchange and the SeqFan mailing list found no proof or counterexample. We therefore classify the result as *new*. The argument is elementary; besides the recurrence of Lemma 3, it uses only Legendre's formula for $v_q(m!)$.

2. Statement

The COMMENTS section of A249434 [14] (entry by Antti Karttunen, November 2, 2014) contains, verbatim:

> `It seems that A006093 gives all those k for which A249151(k) = k. If that is true, then this is a disjoint union of A006093 and A249429.`

Here A006093 [10] is $a(n) = \mathrm{prime}(n) - 1$ ($n \geq 1$), so $k \geq 0$ belongs to A006093 if and only if $k+1$ is prime. A249151 (by Karttunen, October 25, 2014) has the comment "`A006093 seems to give all such k, that a(k) = k.`", A249431 ($a(n) = \mathrm{A249151}(n) - n$) has "`It seems that A006093 gives the positions of zeros.`", and the cross-references of A249434 contain the line "`Subsequences: A006093 (conjectured), A249429, A249430, A249432.`" We read the conjecture as the set equality $\{k \geq 0 : M(k) = k\} = \mathrm{A006093}$, which is Theorem 1; the second sentence of the comment and the cross-reference both presuppose the inclusion $\mathrm{A006093} \subseteq \{k : M(k) = k\}$ as well. That second sentence then follows at once: $\mathrm{A249434} = \{k : M(k) \geq k\}$ is the disjoint union of $\{k : M(k) = k\} = \mathrm{A006093}$ and $\{k : M(k) > k\} = \mathrm{A249429}$.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A249434.lean`. The file was reviewed for fidelity before the proof attempt; that review recorded its SHA-256 hash `638b565a...cb4a`, the file has not changed since, and the hash is listed in `lean/Math0/Statements/FROZEN.sha256`. Its main definitions and theorem statements read as follows (verbatim, in file order; the import, docstrings, blank lines, all proofs, the kernel-friendly helpers `rowProdFact` and `A249151'`, the `DecidablePred` instance and the auxiliary lemmas are omitted, and the two data lists are abbreviated):

```
namespace Math0.Statements.A249434
open Finset Nat
def rowProd (n : ℕ) : ℕ := ∏ j ∈ range (n + 1), n.choose j
theorem rowProd_succ_mul (n : ℕ) : rowProd (n + 1) * (n + 1)! = rowProd n * (n + 1) ^ (n + 1)
def IsTerm (m : ℕ) : Prop := m ! ∣ rowProd m
noncomputable def a (n : ℕ) : ℕ := Nat.nth IsTerm (n - 1)
theorem a_data : (List.range 81).map (fun i => a (i + 1)) = [0, 1, 2, 4, 6, 10, ..., 250, 256]
def lastFact (P : ℕ) : ℕ → ℕ → ℕ
  | 0, m => m
  | f + 1, m => if (m + 1)! ∣ P then lastFact P f (m + 1) else m
def A249151 (n : ℕ) : ℕ := lastFact (rowProd n) (rowProd n) 0
theorem A249151_isGreatest (n : ℕ) : IsGreatest {m : ℕ | m ! ∣ rowProd n} (A249151 n)
theorem A249151_data : (List.range 81).map A249151 = [1, 1, 2, 1, 4, 2, 6, ..., 78, 80, 2]
```

```
def statement : Prop :=
  ∀ k : ℕ, IsGreatest {m : ℕ | m ! ∣ rowProd k} k ↔ (k + 1).Prime
theorem statement_iff : statement ↔ ∀ k : ℕ, A249151 k = k ↔ (k + 1).Prime
end Math0.Statements.A249434
```

`rowProd n` is $P(n)$, and `IsGreatest S k` means that $k \in S$ and k is an upper bound of S. Thus `statement` is Theorem 1, for every $k \in \mathbb{N}$; it does not refer to the search function `A249151`, which implements the OEIS definition by an upward search from $m = 0$. `A249151_isGreatest` proves that this search returns the largest m with $m! \mid P(n)$, and `statement_iff` restates `statement` as $\forall k,\ \mathrm{A249151}(k) = k \leftrightarrow k+1$ prime. The data lemmas compare the definitions with the OEIS by evaluation in the kernel (`decide +kernel`, no `native_decide`): `a_data` checks that the $m < 257$ with $m! \mid P(m)$ are exactly the 81 listed terms of A249434 (the sequence `a` has offset 1), and `A249151_data` checks the 81 listed terms $M(0), \ldots, M(80)$ of A249151. For speed, the kernel computes $(P(n), n!)$ by the recurrence $P(n+1)(n+1)! = P(n)(n+1)^{n+1}$ (`rowProd_succ_mul`, proved in the file). The variable k ranges over all of $\mathbb{N}$, as A249151 has offset 0; for $k = 0$ both sides are false, since $M(0) = 1$ and 1 is not prime.

3. Proof

The proof follows the Lean file `lean/Math0/Frontier/OEIS/A249434.lean`; each step names the corresponding Lean declaration. Throughout, q is a prime and $v = v_q$ is the q-adic valuation of positive integers. For $n \geq 0$ put

$$g(n) = v_q(P(n)), \qquad L(n) = v_q(n!), \qquad S(n) = n - (q-1)L(n)$$

(`g q n` and `L q n` in Lean; S appears only as the expression $n - (q-1)L(n)$). By Legendre's formula $S(n)$ is the sum of the base-q digits of n, but we use only Lemma 8. All identities below are identities between integers; in Lean they are stated in $\mathbb{Z}$ after casting from $\mathbb{N}$.

Lemma 2 (`isGreatest_iff`, `dvd_rowProd_iff`)**.** (a) *For integers $N \geq 1$ and $k \geq 0$, k is the largest m with $m! \mid N$ if and only if $k! \mid N$ and $(k+1)! \nmid N$.* (b) *For integers $d, N \geq 1$, $d \mid N$ if and only if $v_q(d) \leq v_q(N)$ for every prime q.*

Proof. (a) If k is the largest such m, then $k! \mid N$ and, as $k+1 > k$, $(k+1)! \nmid N$. Conversely, if $k! \mid N$ and $m! \mid N$ for some $m > k$, then $(k+1)! \mid m! \mid N$. (b) is unique factorization (Mathlib's `Nat.factorization_prime_le_iff_dvd`). □

Lemma 3 (`rowProd_succ_mul`, `rec_val`)**.** *For $n \geq 0$, $P(n+1)\,(n+1)! = P(n)\,(n+1)^{n+1}$. Consequently, for every prime q,*

$$g(n+1) + L(n+1) = g(n) + (n+1)\,v(n+1). \tag{1}$$

Proof. For $0 \leq j \leq n$ we have $\binom{n}{j}(n+1) = \binom{n+1}{j}(n+1-j)$. Multiplying over $j = 0, \ldots, n$ and using $\prod_{j=0}^{n}(n+1-j) = (n+1)!$ and $\binom{n+1}{n+1} = 1$ gives $P(n)(n+1)^{n+1} = P(n+1)(n+1)!$. Both sides are positive; take v_q. □

Lemma 4 (`key_rel`, `val_fact_iff`, `val_fact_succ_iff`)**.** *Let $k \geq 0$, $n = k+1$ and $D_q(n) = g(n) - (n-1)\,v(n)$. Then $g(k) - L(k) = D_q(n)$. Consequently*

$$k! \mid P(k) \iff D_q(n) \geq 0 \text{ for every prime } q, \tag{2}$$

$$(k+1)! \mid P(k) \iff D_q(n) \geq v_q(n) \text{ for every prime } q. \tag{3}$$

Proof. Since $n! = n \cdot k!$, $L(n) = L(k) + v(n)$; subtracting this from (1) at k gives $g(k) - L(k) = g(n) - k\,v(n)$. Now $v_q(k!) \leq v_q(P(k))$ means $D_q(n) \geq 0$, and $v_q((k+1)!) = v(n) + L(k) \leq g(k)$ means $D_q(n) \geq v(n)$. Apply Lemma 2(b) with $N = P(k) \geq 1$. □

Remark 5. As $P(k)/k! = P(n)/n^{n-1}$ by Lemma 3, Theorem 1 is equivalent to: for $n \geq 1$, $n^{n-1} \mid P(n)$ and $n^n \nmid P(n)$ if and only if n is prime. (This form is not stated in Lean.)

Lemma 6 (`L_mul_add`, `L_mul`, `v_mul`, `v_eq_zero`)**.** *For $m \geq 0$ and $0 \leq r < q$, $L(qm+r) = m + L(m)$. Moreover $v(qm) = 1 + v(m)$ for $m \geq 1$, and $v(qm+r) = 0$ for $0 < r < q$.*

Proof. The multiples of q in $\{1, \dots, qm+r\}$ are $q, 2q, \dots, mq$, whose product is $q^m m!$ (Mathlib's `padicValNat_factorial_mul_add` and `padicValNat_factorial_mul`). The other two claims are clear. □

Lemma 7 (`step_inner`, `block`, `g_mul`, `g_mul_add`, `g_self`)**.** *For $m \geq 0$ and $0 \leq r < q$,*

$$g(qm+r) = q\,g(m) + (q-1-r)\big(m + L(m)\big). \tag{4}$$

In particular $g(q) = q - 1$.

Proof. Within a block. If $r + 1 < q$, apply (1) with $n = qm + r$: here $v(qm+r+1) = 0$ and $L(qm+r+1) = m + L(m)$ by Lemma 6, so $g(qm+r+1) = g(qm+r) - (m + L(m))$. By induction on r,

$$g(qm+r) = g(qm) - r\big(m + L(m)\big) \qquad (0 \leq r < q). \tag{5}$$

The case $r = 0$. We show $g(qm) = q\,g(m) + (q-1)(m + L(m))$ by induction on m. For $m = 0$ both sides vanish, as $P(0) = 0! = 1$. Assume the claim for m. By (5) with $r = q - 1$ and the induction hypothesis, $g(qm + q - 1) = q\,g(m)$. Apply (1) with $n = qm + q - 1$, so that $n + 1 = q(m+1)$, and use $L(q(m+1)) = (m+1) + L(m+1)$ and $v(q(m+1)) = 1 + v(m+1)$ from Lemma 6: $g(q(m+1)) = q\,g(m) + q(m+1)(1 + v(m+1)) - (m+1) - L(m+1)$. By (1) with $n = m$, $q\,g(m) = q\,g(m+1) + q\,L(m+1) - q(m+1)v(m+1)$. Substituting gives $g(q(m+1)) = q\,g(m+1) + (q-1)((m+1) + L(m+1))$, the claim for $m+1$. Combining it with (5) gives (4). Finally, (4) with $m = 1$, $r = 0$ and $g(1) = L(1) = 0$ gives $g(q) = q - 1$. □

Lemma 8 (`S_pos`, `S_two`)**.** (a) *$S(u) \geq 1$ for $u \geq 1$.* (b) *If $u \geq 2$ and $q \nmid u$, then $S(u) \geq 2$.*

Proof. (a) is the inequality $(q-1)v_q(u!) < u$ for $u \neq 0$ (Mathlib's `sub_one_mul_padicValNat_factorial_lt_of_ne_zero`), which follows from Legendre's formula $v_q(u!) = \sum_{i \geq 1} \lfloor u/q^i \rfloor$. (b) Write $u = qm + r$ with $1 \leq r \leq q - 1$; then $L(u) = m + L(m)$. If $m = 0$, then $L(u) = 0$ and $S(u) = u \geq 2$. If $m \geq 1$, then $(q-1)L(m) \leq m - 1$ by (a), so $(q-1)L(u) \leq (q-1)m + m - 1 = qm - 1 \leq u - 2$. □

Lemma 9 (`key_ineq`)**.** *If $g(u) \geq 1$, then $u \leq (q-1)\big(g(u) + L(u)\big)$; that is, $(q-1)\,g(u) \geq S(u)$.*

Proof. By strong induction on u. Write $u = qm + r$ with $0 \leq r < q$, so that $L(u) = m + L(m)$ and $g(u) = q\,g(m) + (q - 1 - r)(m + L(m))$ by Lemma 6 and (4).

Case $r \leq q - 2$. If $m = 0$, then $g(u) = q\,g(0) + (q - 1 - r) \cdot 0 = 0$, contrary to the hypothesis. If $m \geq 1$, then, as $g(m), L(m) \geq 0$, $(q-1)(g(u) + L(u)) = (q-1)\big(q\,g(m) + (q-r)(m + L(m))\big) \geq (q-1)(q-r)m$, and $(q-1)(q-r)m - (qm + r) = mq(q-r-2) + r(m-1) \geq 0$.

Case $r = q - 1$. Then $g(u) = q\,g(m)$, so $g(m) \geq 1$; also $u \geq 1$ (as $g(0) = 0$), hence $m < u$, and the induction hypothesis gives $m \leq (q-1)(g(m) + L(m))$. Therefore

$$\begin{aligned}(q-1)\big(g(u) + L(u)\big) &= (q-1)\big(g(m) + L(m)\big) + (q-1)^2 g(m) + (q-1)m \\ &\geq qm + (q-1)^2 \geq qm + q - 1 = u.\end{aligned}$$

□

Lemma 10 (`L_pow`, `g_pow`)**.** *For $e \geq 0$ and $u \geq 0$,*

$$\begin{aligned}(q-1)L(q^e u) &= (q-1)L(u) + (q^e - 1)u, \\ (q-1)g(q^e u) &= q^e\big((q-1)g(u) - S(u)\big) + (q-1)e\,q^e u + S(u).\end{aligned}$$

Proof. Both by induction on e; the case $e = 0$ is trivial. Put $x = q^e u$. The first identity follows from $L(qx) = x + L(x)$ (Lemma 6). For the second, (4) with $r = 0$ gives $(q-1)g(qx) = q(q-1)g(x) + (q-1)^2 x + (q-1)^2 L(x)$. Insert the induction hypothesis for $(q-1)g(x)$ and the first identity for $(q-1)L(x)$; using $S(u) = u - (q-1)L(u)$, the terms collect to the claimed expression for $e+1$. □

Proposition 11 (`dichotomy`)**.** *Let $n \geq 1$ with $q \mid n$ and $n \neq q$, and let $D = D_q(n) = g(n) - (n-1)v(n)$. Then $D < 0$ or $D \geq v(n)$. More precisely, writing $n = q^e u$ with $q \nmid u$: if $g(u) = 0$ then $D < 0$, and if $g(u) \geq 1$ then $D > e = v(n)$.*

Proof. Write $n = q^e u$ with $q \nmid u$ (Mathlib's `Nat.exists_eq_pow_mul_and_not_dvd`); then $u \geq 1$, $v(n) = e$, and $e \geq 1$ since $q \mid n$. Put $S = S(u) \geq 1$ (Lemma 8). By Lemma 10,

$$(q-1)D = (q-1)g(q^e u) - (q-1)(q^e u - 1)e = q^e\big((q-1)g(u) - S\big) + S + (q-1)e. \tag{6}$$

Case $g(u) = 0$. Then $(q-1)D = (q-1)e - (q^e - 1)S$. If $e = 1$, then $u \neq 1$ because $n \neq q$, so $u \geq 2$ and $S \geq 2$ by Lemma 8(b), and $(q-1)D = (q-1)(1-S) < 0$. If $e \geq 2$, then $(q^e - 1)S \geq q^e - 1 = (q-1)(1 + q + \cdots + q^{e-1}) > (q-1)e$ (`pow_gt`, proved in Lean by induction on e), so again $(q-1)D < 0$. Hence $D < 0$.

Case $g(u) \geq 1$. By Lemma 9, $(q-1)g(u) - S \geq 0$, so (6) gives $(q-1)D \geq S + (q-1)e \geq 1 + (q-1)e$, that is, $(q-1)(D-e) \geq 1$. Hence $D > e$. □

Proof of Theorem 1 (`result`). By Lemma 2(a) we must show: $k! \mid P(k)$ and $(k+1)! \nmid P(k)$ if and only if $n = k+1$ is prime. We use Lemma 4.

n prime. Let q be a prime. If $q \nmid n$, then $v_q(n) = 0$ and $D_q(n) = g(n) \geq 0$. If $q \mid n$, then $q = n$, and $D_n(n) = g(n) - (n-1) \cdot 1 = 0$ by Lemma 7. Hence $k! \mid P(k)$ by (2). Since $D_n(n) = 0 < 1 = v_n(n)$, (3) fails at $q = n$, so $(k+1)! \nmid P(k)$.

n not prime (this includes $n = 1$). Assume $k! \mid P(k)$, so $D_q(n) \geq 0$ for all q by (2). Let q be a prime. If $q \nmid n$, then $D_q(n) \geq 0 = v_q(n)$. If $q \mid n$, then $n \neq q$, so by Proposition 11, as $D_q(n) < 0$ is excluded, $D_q(n) \geq v_q(n)$. By (3), $(k+1)! \mid P(k)$. □

Remark 12 (a description of A249429; not formalized)**.** Let $n = k+1$ be 1 or composite, and for each prime $q \mid n$ let u_q be the part of n prime to q. The proof shows: if $v_q(P(u_q)) \geq 1$ for every prime $q \mid n$, then $(k+1)! \mid P(k)$ and $M(k) > k$; otherwise $D_q(n) < 0$ for some q, and $M(k) < k$. By [5, Theorem 6.7], for $u \geq 1$, $v_q(P(u)) = 0$ if and only if $u = aq^c - 1$ with $1 \leq a \leq q-1$, $c \geq 0$, i.e., $u+1$ has exactly one nonzero base-q digit. Hence, for n not prime, $k \in$ A249429 if and only if for no prime $q \mid n$ does $u_q + 1$ have a single nonzero base-q digit; otherwise $M(k) < k$. For example, $36 = 2^2 \cdot 9 = 3^2 \cdot 4$ with $9 + 1 = 1010_2$ and $4 + 1 = 12_3$, so $35 \in$ A249429. We checked this description numerically for $k < 700$; it is not formalized, and the Lean proof does not use [5].

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A249434.lean` of the math0 repository, https://github.com/Foundation-0/math0. It imports the frozen statement file `lean/Math0/Statements/OEIS/A249434.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is `Math0.Frontier.A249434.result`, of type `Math0.Statements.A249434.statement`. From the statement file the proof uses only `rowProd`, `rowProd_pos`, `rowProd_succ_mul` and `statement`; the data lemmas are not used. Some of the declarations, verbatim (the lemmas inside `section Val` take a variable `(q : ℕ) [hq : Fact q.Prime]`; proofs omitted):

```
private def g (q n : ℕ) : ℕ := padicValNat q (rowProd n)
private def L (q n : ℕ) : ℕ := padicValNat q (n !)
```

```
private lemma dichotomy (n : ℕ) (hn : n ≠ 0) (hdvd : q ∣ n) (hne : n ≠ q) :
    (g q n : ℤ) - ((n : ℤ) - 1) * padicValNat q n < 0 ∨
      (padicValNat q n : ℤ) ≤ (g q n : ℤ) - ((n : ℤ) - 1) * padicValNat q n
theorem result : statement := by
  ...
#math0_settles Math0.Frontier.A249434.result Math0.Statements.A249434.statement
```

All helper definitions and lemmas are `private`. The proof file contains no `sorry`, `admit`, `decide`, `native_decide`, `set_option`, axiom declarations or new syntax; the arithmetic steps are closed by `linarith`, `nlinarith`, `linear_combination`, `ring`, `positivity` and `omega`.

Audit. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a `Prop`-valued definition without universe parameters; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool` from `native_decide`, and custom axioms); and (3) the type of `result` is syntactically the statement constant or its negation, up to metadata, with no unfolding. Here it reports

```
math0 audit: 'Math0.Statements.A249434.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A249434.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A249434.result' proves 'Math0.Statements.A249434.statement'
```

Toolchain and reproduction. We use Lean 4 [7] `v4.33.1` and Mathlib [6] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [4] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command) and checks that the pinned dependencies are unmodified. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A249434.lean` in `lean/`.

References

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Produced by Claude (an AI model developed by Anthropic) in the math0 project